

Arjun Giriraj Day 05

1. What is Ethernet?

A networking technology for Local Area Network, it helps devices to communicate with each other through wire.

2. What is an Ethernet frame?

An Ethernet frame is the unit of data transmitted across ethernet network. Every single unit of ethernet frame contains source address (mac address), destination address (mac address), type, data, and frame check sequence.

3. What is a MAC address?

Media Access Control, it is the physical address of a device. It identifies at layer 2.

4. What does MAC stand for?

Media Access Control.

5. Difference between a MAC address and an IP address?

Mac address is a physical address found on Layer 2 Data Link, IP address is the assigned address on layer 3 Networking.

6. What is a MAC Address Table?

Mac Address Table is a type of record that Switch maintains to map mac address and ports.

7. How does a switch learn MAC addresses?

By receiving the ethernet frame from pc then switch maintains the record in mac address table.

8. What is a source MAC address?

The pc or device's MAC address from which the ethernet frame or data is sent.

9. What is a destination MAC address?

The mac address which receives the ethernet frame or data is destination mac address.

10. What is an unknown unicast frame?

When the destination mac address is not listed in the mac address table so switch floods the frame out of all other ports.

11. What is flooding?

Sending the Frame to all other ports other than source.

12. What is known unicast forwarding?

When the destination mac address is listed in the mac address table so the switch forward that packet in the destined port.

13. At which OSI layer does Ethernet switching primarily operate?

Data link layer, Layer 2

14. Why does a switch maintain a MAC Address Table?

To map all the PC's source address along with their ports, it does easily send the frame afterwards.

15. Why is Ethernet knowledge important in cybersecurity?

Security professionals analyze MAC addresses, Ethernet frames, local traffic, switch behavior, and all as these are the main part to start with Wireshark, Nmap, incident response.

PRACTICAL IDENTIFICATION (4 marks)

Record: Network adapter name, MAC address, IPv4 address, default gateway.

Network adapter name: wlan0, as I use wireless network interface.

MAC Address: 10:6f:d9:7a:da:a9

IPv4 Address: 10.251.71.85

SCREENSHOTS (3 marks)

Attach screenshots of: getmac, ipconfig /all, arp -a

```
venom@VENOM:~$ getmac
10:6f:d9:7a:da:a9
venom@VENOM:~$ ifconfig
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 29236 bytes 2323251 (2.2 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 29236 bytes 2323251 (2.2 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.251.71.85 netmask 255.255.255.0 broadcast 10.251.71.255
    inet6 2401:4900:bcb6:c0a0:f81b:4e11:3751:f1d2 prefixlen 64 scopeid 0x0<global>
    inet6 fe80::83a4:13e:661d:1ec prefixlen 64 scopeid 0x20<link>
    ether 10:6f:d9:7a:da:a9 txqueuelen 1000 (Ethernet)
    RX packets 130250 bytes 108188765 (103.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 67279 bytes 25871505 (24.6 MiB)
    TX errors 0 dropped 76 overruns 0 carrier 0 collisions 0

venom@VENOM:~$ arp -a
gateway (10.251.71.195) at be:b6:a4:e2:f5:64 [ether] on wlan0
```

REPORT QUALITY (3 marks)

Clean formatting, name + Day 05 on page 1, readable screenshots.

